1-zero and BCFW shift

up to cyclic permutations a 1-zero is the imposition of the $n-3$ constraints

$c_{13} = c_{14} = \dots = c_{1,n-1} = 0$ on the kinematic data.

Since $c_{ij} = X_{ij} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}$

the constraints can be equivalently be written as

$$\begin{cases} X_{13} + X_{24} - X_{14} - \underbrace{X_{23}}_{=0} = 0 \\ X_{14} + X_{25} - X_{15} - X_{24} = 0 \\ \vdots \\ X_{1,n-1} + X_{2n} - \underbrace{X_{1n}}_{=0} - X_{2n-1} = 0 \end{cases} \iff \begin{cases} X_{24} = X_{14} - X_{13} \\ X_{25} = X_{15} - X_{13} \\ \vdots \\ X_{2n} = X_{1n} - X_{13} = -X_{13} \end{cases}$$

so the 1-zero fully fixes X_{2j} once X_{1j} is specified, and does not affect X_{ij} if $i,j \notin \{1,2\}$.

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the corresponding BCFW shift is the change $\begin{cases} p_2 \to p_2 + zq \\ p_n \to p_n - zq \end{cases}$

where z is a complex parameter and q satisfies $\begin{cases} q^2 = 0 \\ q \cdot p_2 = q \cdot p_n = 0 \end{cases}$

so that
$(p_2 + zq)^2 = 0$
$(p_n - zq)^2 = 0$

Let $X_{ij}(z)$ be the z-dependent planar invariant.

$X_{ij}(z)$ depends on z only if it includes exactly <u>one</u> of p_2, p_n in the sum, otherwise either there is no z or they cancel out.

$\to$ the only z-dependent ones are $X_{1j}(z)$ $j = 3, \dots, n-1$

and $X_{2j}(z)$ $j = 4, \dots, n$.

Let X_{ij}^{∞} denote the coefficient of the largest power of z appearing in $X_{ij}(z)$.

$$X_{1j}(z) = (P_1 + P_2 + zq + \cdots + P_{j-1})^2 = (P_1 + P_2 + \cdots + P_{j-1})^2 + \underbrace{(zq)^2}_{=0} + 2\, zq \cdot (P_1 + P_2 + \cdots + P_{j-1})$$

$$= X_{1j}(0) + z\; 2q \cdot (P_1 + P_2 + \cdots + P_{j-1}) \quad \Rightarrow \quad X_{1j}^{\infty} = 2q \cdot (P_1 + P_2 + \cdots P_{j-1}) \quad (j = 3, \ldots, n-1)$$

$$X_{2j}(z) = (P_2 + zq + \cdots + P_{j-1})^2 = \cdots$$

$$= X_{2j}(0) + z\; 2q \cdot (P_2 + P_3 + \cdots + P_{j-1}) \quad \Rightarrow \quad X_{2j}^{\infty} = 2q \cdot (P_2 + P_3 + \cdots + P_{j-1}) \quad (j = 4, \ldots, n)$$

in all other cases $X_{ij}(z) = X_{ij}(0)$ so $X_{ij}^{\infty} = X_{ij}(0)$

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note that $X_{2j}^{\infty} = X_{1j}^{\infty} - 2q \cdot P_1 = X_{1j}^{\infty} - X_{13}^{\infty}$ $\quad (j = 4, \ldots, n)$

so the X_{ij}^{∞} satisfy the same constraint as the 1-zero.

$\boxed{\text{Subset zeros}}$

if $\lambda \neq 0$ $\quad B(\lambda X_{ij}) = \lambda^s B(X_{ij})$ for some $s \in \mathbb{Z}$

Let $B(X_{ij})$ be an homogeneous rational function of the X_{ij} variables.

We will write $B(X) = B_m(X) + B_{m-1}(X) + \cdots$

where $B_i(X)$ is the sum of all the terms of B that scales like z^m as $z \to \infty$ under the BCFW shift. (There are finitely many B_i)

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example: $n = 5$, independent X_{ij} are $X_{13}, X_{14}, X_{24}, X_{25}, X_{35}$

scale like z $\qquad$ scales like z^0

if $B = \dfrac{X_{13}}{X_{24} X_{14}} + \dfrac{X_{35}}{X_{13} X_{24}} + \dfrac{1}{X_{35}}$ (note it's homogeneous)

scales like $\dfrac{z}{z^2} = \dfrac{1}{z}$ $\qquad$ scales like $\dfrac{1}{z^2}$ $\qquad$ scales like z^0

then $B_0 = \dfrac{1}{X_{35}}$ $\quad B_{-1} = \dfrac{X_{13}}{X_{24} X_{14}}$ $\quad B_{-2} = \dfrac{X_{35}}{X_{13} X_{24}}$ $\quad$ and $B = B_0 + B_{-1} + B_{-2}$

Def: we say a function $B = B(X_{ij})$ satisfies the 1-zero if

$$X_{2j} = X_{1j} - X_{13} \Rightarrow B(X) = 0 \quad (\forall \text{ choices of } X_{ij} \text{ with } i,j \neq 2)$$

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Prop. The following are equivalent.

① B satisfies the 1-zero

② each B_i satisfies the 1-zero

Proof ② $\Rightarrow$ ① is trivial.

Suppose ① is true. and let $B(z) = B(X_{ij}(z))$

note that as $z \to \infty$ $B(z) = z^m B_m(X_{ij}^\infty) + O(z^{m-1})$

This is because B is a rational function and m is the largest power of z appearing. ex: if

$$B = \underbrace{\frac{X_{13}}{X_{24}X_{14}}}_{B_{-1}} + \underbrace{\frac{X_{35}}{X_{13}X_{24}}}_{B_{-2}}$$

then

$$B(z) = B_{-1}(z) + O(z^{-2})$$

$$= \frac{X_{13}(0) + z X_{13}^\infty}{(X_{24}(0) + z X_{24}^\infty)(X_{14}(0) + z X_{14}^\infty)} + O(z^{-2})$$

$$= \frac{1}{z} \frac{X_{13}^\infty}{X_{24}^\infty X_{14}^\infty} + O(z^{-2}) = z^{-1} B_{-1}(X_{ij}^\infty) + O(z^{-2})$$

since $X_{2j}^\infty = X_{1j}^\infty - X_{13}^\infty$

note that if $X_{2j}(0) = X_{1j}(0) - X_{13}(0)$ then $X_{2j}(z) = X_{1j}(z) - X_{13}(z)$ $\forall z$

so $B(z)$ satisfies the zero $\forall z$. But then

$$0 = z^m B_m(X_{ij}^\infty) + O(z^{m-1}) \quad \forall z \quad \text{so it must be} \quad B_m(X_{ij}^\infty) = 0$$

the only constraints on the X_{ij}^∞ are the same as the 1-zero, so this means B_m satisfies the 1-zero.

Then we proceed by induction. if B satisfies the 1-zero and B_m satisfies the 1-zero then $B' = B - B_m = B_{m-1} + B_{m-2} + \dots$ satisfies the 1-zero. Following the same steps as above we then prove that B_{m-1} satisfies the 1-zero and so on.